SQL Practice Exercise 4: Join, Union, Filtering, and Aggregates

Bright Light Tutorials

Table 1: Employees

EmployeeID	FirstName	LastName	Department	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Table 2: Projects

ProjectID	EmployeeID	ProjectName	Budget	StartDate	Status
101	1	AI Development	100000	2024-01-10	Completed
102	2	Employee training	50000	2024-02-15	Ongoing
103	1	Cybersecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

Questions

1. SQL Joins

1.1 Retrieve all employees and their assigned projects using an INNER JOIN.

```
Select
        EmployeeID,
        FirstName,
        LastName,
        Department,
        Salary, ProjectID, ProjectName,
        Budget, Status
From Employees AS A
INNER Join Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

-- Return Results:

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersec Audit	75000	pending
2	Alice	Smith	HR	60000	102	Employee training	50000	ongoing
3	Bob	Johnson	Finance	75000	104	Finance Analysis	90000	ongoing
5	EMMa	Wilson	Sales	65000	105	Market Expansion	65000	completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	pending

1.2 Retrieve all employees and their assigned projects including employees who have no projects using a LEFT JOIN

```
Select EmployeeID, FirstName, LastName, Department, Salary,
        ProjectID, ProjectName, Budget, Status
From Employees AS A
LEFT Join Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

-- Return Results

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	completed
1	John	Doe	IT	70000	103	Cybersec Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee training	50000	ongoing
3	Bob	John	Finance	75000	104	Finance Analysis	90000	ongoing
4	David	Brown	IT	72000	null	null	null	null
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

1. Retrieve all employees and projects, including those without a match in either table using a FULL OUTER JOIN.

```
Select EmployeeID, FirstName, LastName, Department,
       Salary, ProjectID, ProjectName, Budget, Status
From employees AS A
FULL OUTER JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;
```

-- Return Results:

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Dev	100000	Completed
1	John	Doe	IT	70000	103	Cyber Security	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	ongoing
4	David	Brown	IT	72000	Null	Null	Null	Null
5	Emma	wilson	Sales	65000	105	Market Expansion	65000	completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

3. Filtering Statements

7. Retrieve Employees who earn More than 70000.

```
Select EmployeeID, FirstName, LastName,
       Department, Salary
From Employees
where Salary > 70000;
```

-- Return Results

EmployeeID	FirstName	Department	Salary
3	Bob	Finance	75000
4	David	Brown	72000
6	Michael	Clark	80000

8. Retrieve employees working in either IT or Finance departments.

--

Select EmployeeID, FirstName, LastName, Department, Salary
From Employees
Where department = 'IT'
or department = 'Finance';

-- Return Results:

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. Retrieve projects that are not yet completed

Select ProjectID, ProjectName, Budget, Status
From Projects
where Status not in (completed);

-- Return Results

ProjectID	ProjectName	Budget	Status
102	Employee training	50000	ONGOING
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	ONGOING
106	Risk management	80000	Pending

10. Retrieve projects that have a budget greater than 70000 and are not completed.

```
Select ProjectID, ProjectName, Budget, Status
From projects
where budget > 70000
and Status not in (completed);
```

-- Return Results

ProjectID	ProjectName	Budget	Status
103	Cyber security Audit	75000	Pending
104	Financial Analysis	90000	ongoing
106	Risk Management	80000	Pending

11. Retrieve employees from New York OR Toronto, ordered by Salary in descending order.

```
Select EmployeeID, FirstName, LastName, Department,
       Salary, City
From employees
where City = 'New York' or City = 'Toronto'
order by Salary desc;
```

-- Return Results

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12. Retrieve the top 3 highest-paid employees.

```
Select EmployeeID, FirstName, LastName, Department,
       Salary
From employees
order by Salary desc
limit 3;
```

-- Return Results

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

13.
4. Aggregate Functions with Group By & HAVING

13. Find the total salary per department, sorted in descending order.

```
Select Department,
        sum(Salary) AS Total_Salary
From employees
group by Department
order by salary desc;
```

Return Results:

Department	Total_Salary
Finance	155000
IT	142000
Sales	65000
HR	60000

14. Find the average salary per city, but only include cities where the average salary is greater than 65000.

```
Select city,
        AVG(salary) AS Average_Salary
From employees
group by city
having salary > 65000;
```

Return Results:

City	Average_Salary
New York	80000
Toronto	75000
London	72000
New York	70000

15. Count the number of employees per department, including only departments with more than 1 employee.

```
Select Department,
        Count(employeeID) AS Employee_Count
From employees
group by department
having employeeID > 1;
```

Return Results:

Department	Employee_Count
IT	2
Finance	2

16. Retrieve the number of projects per status, but only include statuses with atleast 2 projects.

```
Select Status,
       Count(ProjectID) AS Project_Count
From projects
group by Status
having projectID = 2;
```

-- Return Results:

Status	Project_Count
Completed	2
Ongoing	2
Pending	2

17. Retrieve the total Project budget per employee, but only for employees who are managing projects worth more than 150 000.

```
Select EmployeeID,
       FIRSTname,
       Lastname,
       Sum(Budget) AS Total_Project_Budget
From Projects
group by ~~budget~~ EmployeeID
having budget > 150 000;
```

-- Return Results:

EmployeeID	FirstName	LastName	Total_Project_budget
1	null	null	175 000